



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2024/2025

THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE IN AGRONOMY,
BACHELOR OF SCIENCE IN ANIMAL SCIENCE, BACHELOR
OF SCIENCE IN ENVIRONMENTAL SCIENCE, BACHELOR
OF SCIENCE IN AGIBUSINES MANAGEMENT, BACHELOR
BACHELOR OF SCIENCE IN HORTICULTURE WITH
INFORMATION TECHNOLOGY

MAIN CAMPUS

FIT 303: GEOGRAPHIC INFORMATION SYSTEM

Date: 3rd February, 2025

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Do not write anywhere on this Question paper
- Answer Question ONE and any other THREE.

QUESTION ONE

- a. Explain the components of GIS as used in Agriculture and Environment (5marks)
- b. Explain the functions of GIS in solving the current environmental challenges (6marks)
- c. A layered architecture for a spatial database management system is composed of hierarchy of modules. Explain different layers of a spatial database management (10marks)
- d. Discuss vector data application in environmental conservation (4 marks)

QUESTION TWO

Discuss the criteria used for classifying map projection and state the major characteristics of each type of projection. (15marks)

QUESTION THREE

- a. Explain sources of attribute data used in GIS
- b. Inconsistent data can have a significant impact on any organization's success as inaccurate information may jeopardize safety. Explain different types of spatial data input errors and how they can be eliminated (10 marks)

QUESTION FOUR

- a. Compare the hierarchical model with the relational model in terms of ease of learning and ease of use. (7marks)
- b. The ability of Object-oriented database models to handle complex data efficiently and align with modern programming practices positions them as a critical technology for future data-driven applications. Explain the real-world applications of Object-oriented database models (8 marks).

QUESTION FIVE

Geographic Information Systems are useful in various domains. Explain the various applications of raster data and vector data in agriculture and environment. (15 marks)

QUESTION SIX

The ministry of agriculture wants to publish information about weather conditions in January and February. They have temperature, rainfall and wind strength data. Discuss the type of analysis you will use to estimate places that will have optimal weather conditions with mild temperatures, no rainfall and little wind strength and identify areas in your region that meet these criteria. (15 marks)